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Auditing Collapse Narratives: An Integrated Framework for Cost-Sensitive Threshold Calibration and Hex-Encoded Discourse Analysis

This report provides a comprehensive deep-dive analysis into the development and implementation of a configurable threshold-sweep script for the GESP claim-type classifier. It details the architecture of the Lua-based tool, the multi-objective optimization framework that prioritizes safety-critical objectives, and the innovative integration of a hex-encoded weighted doom/agency (dD) decoder. The analysis covers the systematic joint calibration of claim-type rules and dD bands, the mechanisms for targeted error analysis and domain robustness evaluation, and the mathematical foundations of the weighted decoding process. Finally, it explores how the structured, hex-coded outputs facilitate the creation of an auditable research environment through error-tensor dashboards and trajectory analysis, providing a complete blueprint for trustworthy and principled computational analysis of complex narrative structures.

The Integrated Sweep Script Architecture and Multi-Metric Optimization

The core of the GESP research initiative is a sophisticated Lua-based sweep script, `gesp_threshold_sweep.lua`, designed not merely as a tuning utility but as a comprehensive validation and optimization engine [1]. Its architecture is engineered around the principle of modularity and separation of concerns, systematically exploring a defined parameter space to find an optimal set of thresholds that balances multiple, often competing, performance objectives. The script orchestrates a pipeline that iterates over a grid of hyperparameters, configures the analytical stack accordingly, runs the classifier against a labeled gold corpus, computes a suite of diagnostic metrics, and makes a data-driven decision on the best-performing configuration [41] [42]. This approach moves beyond simple accuracy-focused machine learning, embedding principles of safety engineering and cost-sensitive evaluation directly into the model selection process.

The script's primary function is to define and explore the parameter space for two distinct but interconnected sets of variables: claim-type classification thresholds and globally consistent dD band boundaries [60]. The claim-type thresholds—`t_dom` for dominant dimension cutoff, `t_all` for all-dimensions-high cutoff, and `t_doom` for doom-saturation cutoff—are fundamental to the rule-based logic that classifies a snippet into one of five categories, including Single-Factor Doom and Everything-Is-Broken [61]. The dD band boundaries, represented by `dD_low` and `dD_high`, define the quantiles that segment the continuous doom/agency metric into qualitative labels like "agency-leaning," "mixed," "doom-leaning," and "doom-saturated" [67]. By jointly tuning these parameters, the script acknowledges their inherent interdependence; for instance, the classification of a snippet as "Single-Factor Doom" is contingent upon whether its dD value exceeds a certain threshold [45]. The script formalizes this relationship by defining a Cartesian product of these parameters within a `default_theta_grid()` function, which generates all possible combinations for exhaustive search [65].

A critical architectural feature is the centralization of configuration. The script does not hard-code any thresholds or band values into the classifier's logic. Instead, it exposes them through a configuration object passed to the relevant modules [1]. The `configure_stack_with_theta(theta)` function serves as the nexus for this orchestration, updating both the `ClaimClassifier.thresholds` table and the `Debias.dDbands` table simultaneously [66]. This ensures that when a new combination of parameters (a single theta from the grid) is tested, the entire downstream analytical pipeline—including the Debias analyzer and the ClaimClassifier—sees a self-consistent state. This design prevents subtle bugs and logical inconsistencies that would arise from partial or out-of-sync updates, forming the foundation for reliable and reproducible calibration [68]. The unique identity of each tested configuration is captured by the `theta_id(theta)` function, which formats the parameter values into a human-readable string like `tdom=0.50_tall=0.70_tdoom=0.60_dlow=0.35_dhigh=0.65`, providing a clear audit trail of every experiment run [64].

The evaluation itself is multi-faceted, reflecting the project's dual focus on classification accuracy and regression precision. For each candidate `theta`, the script executes a full pass over the gold corpus, generating predictions (`sys`) alongside the ground-truth labels (`gold`). From these paired results, it calculates a rich set of metrics. At the classification level, it computes the macro-F1 score, which averages the F1 scores for each of the five claim types, providing a balanced measure of overall performance [13]. It also generates a full confusion matrix, allowing for detailed error analysis by visualizing which classes are being confused with others [30]. Concurrently, at the regression level, it calculates the

Mean Absolute Error (MAE) for the dD metric, measuring how closely the predicted doom/agency ratio aligns with the annotated value [72]. This MAE is a direct indicator of the quantitative accuracy of the underlying discourse analysis before it even reaches the claim-type classification stage.

The most distinctive aspect of the evaluation framework is its explicit incorporation of a cost-sensitive loss function. This function moves beyond standard classification metrics to penalize specific types of errors based on their perceived severity. The default configuration of the script assigns stronger penalties to high-risk failure modes, specifically false negatives on snippets with high-stress in societal (S) or personal (P) dimensions, and false positives where doom-saturated discourse is mislabeled as less severe [47]. This reflects a "safe-by-design" philosophy, where the primary goal is to minimize the risk of underestimating existential threats rather than simply maximizing raw accuracy [10] [14]. The optimization objective is formally expressed as a composite score:
J = λ₁ \cdot \text{macro-F1} + λ₂ \cdot (1 - \text{MAE}<em _text_high-risk="\text{hig
[1] [16]. In this formulation, minimizing the cost term (due to the negative sign) is a primary directive. The user's clarification specifies that while the trade-off between these terms should be configurable via the lambda weights (λ_1,λ_2,λ_3), the shipped default configuration must heavily favor the cost-sensitive objective, ensuring the tool's immediate use aligns with safety priorities [60]. The final selection of the best }`theta` is governed by a clear decision rule: prefer the configuration with the lowest cost; if costs are tied, break the tie by selecting the one with the highest macro-F1 score [41]. This prioritizes safety first, then general classification performance. The output of the script is a comprehensive JSON file containing the selected `best_theta`, the rationale for its selection ("min_cost_then_max_macroF1"), the full performance summary for that configuration across the entire corpus and per-domain splits, and the corresponding confusion matrix [2].

Targeted Error Analysis and Domain-Specific Robustness Assessment

A key strength of the proposed threshold-sweep framework is its ability to look beyond aggregate performance metrics and perform targeted analyses on specific, high-value subsets of the data. This capability directly addresses the need for iterative improvement of annotation guidelines and ensures the resulting classifier is not just statistically

accurate but also practically useful and robust across diverse contexts. The script is explicitly designed to isolate and evaluate performance on two critical types of cases: items where human annotators disagreed and instances involving the confusable pair of claim types, "Moral Panic" versus "Everything-Is-Broken" [30] [31].

The process begins during the data loading phase, where the script calls dedicated helper functions to partition the gold corpus. The `select_disagreement_subset(records)` function filters the records to include only those snippets for which annotators provided disagreement notes, signaling ambiguity in the ground truth [32]. Similarly, the `select_moralpanic_vs_eib(records)` function creates a subset containing all snippets classified as either "Moral Panic" or "Everything-Is-Broken" [35]. These specialized subsets are then fed into separate evaluation routines. For each candidate `theta` in the sweep, the script runs a targeted error analysis function, `targeted_error_analysis()`, on these subsets. This function re-runs the classifier and compares the predictions against the ground truth, but its focus is narrower. For the disagreement subset, it tracks the rate at which the classifier correctly identifies the correct label, effectively measuring how well the rules resolve ambiguity. For the Moral Panic vs. EIB subset, it specifically counts how many of these confusable cases are resolved correctly under the current `theta`. The result is a detailed breakdown of how each configuration impacts these crucial error types, logging metrics like the number of fixed errors and the fix rate [34]. This granular feedback loop is invaluable; researchers can inspect which `theta` values successfully resolve specific, known-confusion cases and use this evidence to refine the underlying rules or update the annotation guidelines to reduce future ambiguity [35]. This transforms calibration from a black-box optimization problem into a transparent process of guided discovery.

Beyond targeted error analysis, the framework places a strong emphasis on evaluating the robustness of the selected thresholds across different domains of discourse. The assumption that a single set of rules will perform optimally everywhere is a common pitfall in NLP. To counteract this, the script includes a domain-specific robustness evaluation module [23]. Upon loading the corpus, the `split_by_domain()` function categorizes each record into 'fiction', 'news', or 'social' based on context source metadata [46]. After evaluating a given `theta` on the entire corpus (the 'all' split), the script recomputes its performance metrics—specifically macro-F1—on each of these three domain-specific splits separately [68]. This allows for a direct comparison of performance across genres and platforms.

The strategic response to detecting performance degradation is carefully specified. The script is designed to *flag* any significant drop in performance in its output summary rather than automatically switching to a different, domain-specific configuration [60]. This conservative approach preserves the integrity of the main global configuration, preventing silent divergence and maintaining simplicity. The flagging mechanism might involve comparing the per-domain macro-F1 scores to the global score and highlighting any that fall below a predefined tolerance threshold (e.g., a >5% decrease). Alongside the flags, the script can optionally generate suggestions for domain-specific threshold variants, but these would be presented as recommendations for manual review rather than being applied automatically [60]. This workflow ensures that any decision to adopt specialized rules for a particular domain is deliberate and justified. The final output of the sweep includes a `domains` object within the best-theta summary, containing the full evaluation results for each domain, providing a complete picture of the model's behavior in different contexts [1]. This commitment to cross-domain robustness assessment is a hallmark of building systems intended for real-world deployment, where input data will inevitably come from a variety of sources and styles [29].

Feature	Description	Purpose
Disagreement Subset Analysis	Identifies snippets with annotator disagreement notes.	Evaluates how well the classifier resolves ambiguous cases.
Moral Panic vs. EIB Analysis	Isolates snippets classified as either "Moral Panic" or "Everything-Is-Broken".	Measures the model's ability to distinguish between these confusable, high-stakes claim types.
Targeted Error Metrics	Tracks the number and rate of resolved errors for specific target subsets.	Provides granular feedback for improving rules and annotation guidelines.
Domain Splitting	Partitions the corpus into fiction, news, and social media subsets.	Enables evaluation of model performance and potential biases across different text genres.
Robustness Flagging	Compares per-domain performance (macro-F1) against global performance.	Alerts researchers to potential overfitting to a specific domain without automatic intervention.

Hex-Encoded Weighted Doom/Agency Decoding and Calibration

The second major pillar of the GESP research framework is the development of a weighted doom/agency (dD) decoder, which represents a significant evolution from a simple, unweighted token-counting method. This decoder introduces linguistic nuance by assigning variable weights to tokens based on their semantic role, such as intensifiers, absolutes, and hedges, thereby creating a more sensitive and context-aware measure of

discourse tone [3] [5]. The innovation lies not just in the weighted calculation itself but in its formalization into a hex-encoded object, complete with a versioning scheme, that enables precise, auditable experimentation and comparison. This transforms the decoder from a heuristic tweak into a rigorously defined and documented research instrument.

The core concept behind the weighted decoder is to model the intensity of doom and agency expressed in a snippet more faithfully than a unit-weight count could. Instead of treating every doom-related word equally, the decoder uses lexicons to assign higher weights to words that amplify negativity (e.g., "guaranteed," "never") and lower weights or polarity-flipping effects to words that hedge or negate it (e.g., "might," "not doomed") [8] [85]. The final weighted counts for doom and agency are then calculated by summing these token-level weights. The weighted doom/agency ratio, `d^w`, is then computed as the ratio of total weighted doom tokens to the sum of weighted doom and agency tokens [89]. This richer signal is expected to better capture the subtleties of rhetorical strategies used in collapse narratives, leading to improved accuracy in predicting both the continuous dD metric (reducing MAE) and the discrete claim types that depend on it.

The most novel aspect of this framework is its systematic use of hexadecimal encoding to manage the complexity of the decoder's configuration and output. The decoder's operational mode is defined by a series of hyperparameters governing intensifiers, hedges, negation handling, and normalization. Each of these four regimes is mapped to a single nibble (4-bit value), creating a compact, 4-nibble hexadecimal identifier, such as `0xIHNN` [90]. The `I` nibble encodes the range of intensifier multipliers used, the `H` nibble encodes the hedge multiplier strategy, and the `NN` nibbles encode the specifics of the negation scope and flip/attenuation rules [67]. This `0xIHNN` code acts as a machine-readable fingerprint for each experimental variant of the decoder. Researchers can precisely replicate experiments, compare the effects of changing a single component (e.g., swapping a negation strategy), and maintain a clear history of which decoder version produced which results, a critical feature for reproducible and trustworthy scientific practice [59].

The calibration of these decoder parameters is driven by empirical optimization on the gold corpus. Rather than relying on arbitrary choices, the weights and rules are tuned to minimize a combined objective function that includes both regression error and cost-sensitive loss [72] [87]. Specifically, the parameters are optimized to minimize the Mean Absolute Error (MAE) on the dD prediction task and to reduce the cost-sensitive loss associated with mislabeling doom-saturated discourse, especially in high-stakes S and P dimensions [47]. This creates a powerful feedback loop: the quantitative performance on the gold data directly informs the qualitative aspects of the decoder's linguistic sensitivity.

Once a decoder configuration has been empirically validated against the gold subset, demonstrating sufficient improvements in MAE and cost-loss, its maturity is marked with a "Usefulness Status Rating" (USR) hex tag [1]. Partially calibrated decoders might reside in a `0x8Cxx` range, while fully validated, mature versions are promoted to a `0x9Fxx` range [60]. This USR tag, potentially with trailing nibbles encoding binned performance gains (e.g., ΔMAE, ΔL), provides a quick, verifiable status indicator of the decoder's reliability for downstream tasks [68]. This structured, evidence-based approach to calibration and versioning elevates the decoder from a simple preprocessing step to a first-class, auditable research object.

Mathematical Formulation and Parameterization of the Weighted Decoder

The transition from an unweighted to a weighted doom/agency (dD) metric is grounded in a set of explicit mathematical formulations that define how linguistic features are translated into a quantitative score. These formulas provide the theoretical basis for the decoder's operation and enable its rigorous calibration and hex-based parameterization. The core of the model involves modifying the base weight of each token based on its linguistic modifiers before aggregating them to compute the final doom/agency ratios.

Let each token t in a snippet possess a base label $b(t) \in \{doom, agency, other\}$ and a base weight $w_0(t)$, which is 1 if the token is a doom or agency term and 0 otherwise [90]. The effective weight $w'(t)$ of the token is then modified by several factors. First, an intensifier multiplier $m_i(t) \geq 1$ is applied if the token is an intensifier co-occurring with a doom term (e.g., mi might be 1.5 for "certain," up to a maximum $m_{\beta ax}$) [3]. Second, a hedge multiplier $m_h(t) \in (0,1]$ is applied to agency hedges or to attenuate doom under hedged statements (e.g., mh might be 0.5 for "might not," down to a minimum $m_{\beta in}$) [8]. Third, a negation operator N comes into play. If a doom token falls within the scope of a negator (like "not" or "never"), its effective weight is altered. This can be a full flip, turning doom into agency, or an attenuation, reducing its weight towards zero [85]. The final, signed weight $w'(t)$ incorporates these factors.

The weighted counts are then calculated by summing these adjusted weights over all tokens in the snippet. The total weighted counts for doom and agency are:

N^{w}<em : b_t_="\text{doom" t>{\text{doom}} = \sum}} w'(t), \quad N^{w

These global counts are used to compute the overall weighted doom ratio:

d^{w} = \frac{N^{w}}<em _text_doom="\text{doom">{\text{doom}}}{N^{w}</em

If dimension hints are available (e.g., a token is specifically linked to societal or personal stress), these can be used to compute per-dimension weighted counts and ratios, }} + N^{w}_{\text{agency}}N^{w}_{D,\text{doom}} and d^{w}_D, following the same logic [89]. This structure mirrors the original unweighted formula but replaces simple counts with weighted sums, making the final metric more responsive to the rhetorical force of the language.

To translate these parameterized models into the hex-encoded framework, each component's parameter range is quantized into a 4-bit nibble. For example, the intensifier multiplier m_i is mapped to a nibble v_i using a linear scaling formula:

$$v_I = \left\lfloor 15 \cdot \frac{m_i^{\text{doom}} - 1}{m_i^{\max} - 1} \right\rfloor$$

This maps the multiplicative factor above 1 down to a value from 0 to 15, where 0 might mean no intensification and 15 means maximal intensification. A similar formula is used for the hedge multiplier m_h, mapping factors below 1 to a nibble v_H:

$$v_H = \left\lfloor 15 \cdot \frac{1 - m_h^{\text{agency}}}{1 - m_h^{\min}} \right\rfloor$$

Other hyperparameters, such as the window size for negation scopes or smoothing constants for division by zero, are similarly quantized into the remaining nibbles of the 0xIHNN configuration hex [1]. This mapping allows any point in the high-dimensional parameter space to be collapsed into a simple, 16-bit integer, creating a unique and interpretable ID for each decoder variant.

The calibration process itself becomes a constrained optimization problem. The goal is to find the parameter vector θ (which defines the decoder configuration) that minimizes a combined loss function on the gold corpus. This loss function typically balances the reduction of regression error (MAE) with the reduction of cost-sensitive loss (L):

$$\min_{\theta} \alpha \cdot \mathrm{MAE}_{dD}(\theta) + (1-\alpha) \cdot \frac{L(\theta)}{L_0}$$

Here, α is a balancing coefficient, MAE_{dD}(θ) is the Mean Absolute Error of the weighted dD prediction, L(θ) is the cost-sensitive loss, and L_0 is a normalization constant, often the baseline loss of an unweighted model [72]. The optimization is subject to constraints on the parameter ranges to ensure stability and interpretability. Solutions to this optimization problem, θ*, are then discretized back into the 0xIHNN hex format, completing the loop from abstract parameters to concrete, auditable identifiers [1]. This mathematical rigor ensures that the decoder's evolution is data-driven and transparent.

Component	Mathematical Formula
Token Weighting	$w'(t) = w_0(t) \cdot m_i(t) \cdot m_h(t)$ (with negation flip/attenuation) [90]
Global Doom Ratio	d^{w} = \frac{N^{w}<em _text_doom="\text{doom">{\text{doom}}}{N^{w}
Per-Dimension Ratio	d^{w}D = \frac{N^{w}}}}{N^{w{D,\te [89] }}
Decoder Config	Quantization of parameter ranges (e.g., $m_i^{\max}, m_h^{\min}$) [1]
Band-Shift Flag	Discrete change in qualitative band label (e.g., mixed → doom-leaning) [1]

Structured Outputs and the Error-Tensor Dashboard for Auditable Research

The true power of the integrated GESP framework emerges from the structured, hex-encoded outputs generated by the weighted dD decoder and the comprehensive reporting of the threshold-sweep script. These outputs transform a collection of numerical metrics into a rich, auditable dataset that can be used for advanced diagnostics, visualization, and long-term monitoring of the system's "health" [68]. By packaging the essential information about a snippet's analysis into compact, decodable strings, the framework enables researchers to ask and answer complex questions about discourse dynamics, error patterns, and the effectiveness of interventions that would be intractable with traditional, unstructured logs.

For each snippet processed, the system produces a multi-layered output signature. The most important is the **per-snippet doom-agency state**, which is encoded into a hex string like `0xD GESP S` [1]. This string is a compact representation of the weighted analysis. The first nibble (`D`) encodes the global weighted doom ratio `d^w` into a 4-bit integer (`⌊15 · d^w⌋`). Following this is a `0x` prefix and the GESP-style tag, where the nibbles for G, E, S, and P encode the respective per-dimension ratios (`d^w_G`, `d^w_E`, `d^w_S`, `d^w_P`) [67]. Finally, the last nibble (`S`) is a band-shift flag, which captures the effect of the weighted analysis compared to a baseline (e.g., unweighted) decoder. A value of `0xE` indicates the weighted analysis pushed the snippet's qualitative band to a more doom-laden category, `0x1` indicates a shift to a less doom-laden category, and `0x0` signifies no change [68]. This single string contains a wealth of information: the nuanced, weighted tonal profile of the text, and the degree to which this profile differs from a simpler interpretation.

This structured output is the key ingredient for constructing an **error-tensor dashboard**. An error tensor is a multi-dimensional array that stores performance metrics for every combination of factors that could influence an error. The GESP framework extends this concept by creating a tensor indexed by dimension (D), claim type (c), analogy pattern (a), and now, importantly, decoder configuration (θ), identified by its `0xIHNN` hex code [1] [59]. For each cell in this four-dimensional space, the system can compute and store metrics like the average MAE on dD, the cost-sensitive loss, and the claim-type F1. For instance, one could examine the error profile for all snippets tagged with the `violenceequalscollapse` analogy that were classified as "Moral Panic" and processed by the `0x8C5A` decoder. By mapping these error metrics (e.g., MAE, cost) to a color spectrum using hex codes (e.g., low error = 0x1-0x3, high error = 0xC-F), this vast

dataset can be rendered as a "heat map" or a small Markdown table [1]. This visualization immediately highlights high-error regions, guiding future research efforts. For example, if the dashboard shows consistently high cost-loss for the slice `(D=S, c=Scholarly Risk, θ=0x8C5A)`, it signals that the current decoder is particularly prone to costly mistakes when analyzing scholarly texts employing a specific rhetorical pattern. This directs researchers to either expand the corpus with more examples from this region or to tune the decoder's parameters for this specific context [1].

Furthermore, these structured outputs enable **trajectory and intervention studies**. By tracking the sequence of doom-agency hex states (`0xD GESP S`) over time in a conversation thread or a workshop, researchers can quantify shifts in collective discourse [68]. They can measure whether a specific intervention, such as the introduction of a playbook prompt, correlates with a systematic shift in the band-shift flag `S` from positive (more doom) towards negative (less doom), indicating a successful "de-dooming" effect [1]. Crucially, this can be done while monitoring the underlying stress (`sclaimed`) and distortion (`distortion`) metrics to ensure that a reduction in doom perception is not simply a case of denial (where `D` remains high but `d` drops) [1]. The outcome of such a study could be summarized with another hex signature, like `0xTSRD`, which encodes the trend in the S-band, the final qualitative band, the change in distortion magnitude, and the domain [1]. This turns qualitative assessments of discourse health into quantitative, repeatable measurements. Ultimately, by documenting the fields for the decoder config hex, doom-agency hex, and decoder USR hex in the corpus schema and metrics specification, the framework establishes a persistent, versioned registry of its own analytical state, ensuring that all future work is built upon a solid, transparent, and auditable foundation [1].

Synthesis and Strategic Implications for Trustworthy AI

The development of the configurable threshold-sweep script and the hex-encoded weighted dD decoder constitutes a holistic and principled approach to building a trustworthy and robust system for analyzing collapse-related narratives. The project's strategic significance lies not in the incremental improvement of a single metric, but in the construction of an integrated, auditable, and safety-conscious research ecosystem. It embodies a "primitives-first" methodology, where foundational, quantifiable metrics for discourse analysis (dD MAE) and classification (macro-F1, cost-loss) are established

before higher-level claims are made [1]. This ensures that the entire analytical stack is grounded in empirical evidence rather than subjective interpretation.

The core innovation of the framework is its multi-objective optimization, which explicitly prioritizes safety by default. The use of a configurable, cost-sensitive loss function that heavily penalizes underestimation of high-stakes risks (e.g., missing doom-saturated S/P discourse) embeds a "guardrails-first" philosophy into the model selection process [47] [60]. This is a critical design choice for any system operating in the domain of existential risk analysis, where the consequences of failure are non-trivial. The configurability of the trade-off via lambda weights provides a pragmatic balance, allowing the system to be repurposed for different research questions while maintaining its default alignment with safety goals. This demonstrates a mature understanding of the tension between statistical performance and ethical responsibility in AI.

The joint calibration of claim-type rules and dD bands is another cornerstone of the framework's robustness. By recognizing the deep interdependencies between these components and ensuring they are tuned together as a coherent system, the framework avoids the suboptimal solutions that arise from treating them in isolation [40]. The weighted dD decoder further refines this system by introducing linguistic nuance, moving beyond simplistic keyword matching to a more sophisticated analysis of rhetorical force. The adoption of a hex-based encoding scheme for the decoder's configuration (`0xIHNN`), its outputs (`0xD GESP S`), and its calibration status (`0x9Fxx`) is a masterstroke of engineering for reproducibility and auditability [1]. It transforms a complex, high-dimensional parameter space into a series of simple, machine-readable identifiers, enabling precise tracking of experimental variants and their consequences. This level of transparency is essential for building trust in the system's outputs and for facilitating collaborative research.

Finally, the framework's capacity for targeted analysis and structured reporting empowers a cycle of continuous improvement. The ability to drill down into specific error patterns, such as annotator disagreements or confusions between "Moral Panic" and "Everything-Is-Broken," provides direct feedback for refining both the model's rules and the annotation guidelines [30]. The error-tensor dashboard, built from the structured outputs, serves as a powerful diagnostic tool, illuminating weaknesses in the system's performance across a vast, multi-dimensional space of inputs. This data-driven approach to identifying and addressing weaknesses is far superior to ad-hoc debugging. In synthesis, this research agenda provides a complete and forward-looking blueprint for computationally rigorous, ethically aligned, and auditable analysis of complex human narratives. It successfully integrates cutting-edge ideas from cost-sensitive learning, sentiment analysis, and

software engineering practices to create a system that is not only accurate but also interpretable, traceable, and aligned with its ultimate purpose of enhancing our understanding of societal risks.

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